ONLINE SHOPPERS PURCHASING INTENTION

Dataset Knowledge Base for RAG and Streamlit

Purpose of this document

This knowledge base explains the Online Shoppers Purchasing Intention dataset. It focuses on the dataset, its columns, the meaning of recorded session information, and common questions a user may ask through a dataset chatbot. It is not a general machine-learning textbook.

1. Dataset overview

• **Dataset name:** Online Shoppers Purchasing Intention Dataset.

• **What one row represents:** One browsing session on an online shopping website.

• **What the dataset describes:** Page visits, time spent on page categories, technical/session attributes, and whether a purchase occurred.

• **Target column:** Revenue. In the original dataset, True means the session resulted in a purchase; False means it did not.

• **Original dataset size:** 12,330 sessions and 18 columns.

• **Local project file:** The previously cleaned project CSV contained 12,205 rows after cleaning. This is the processed local dataset, not the original dataset size.

2. How to read one row

A row combines browsing behaviour (for example, counts of product-related pages and time spent on them) with session details such as month, visitor type, browser, operating system, region, traffic type, and weekend. Revenue records the observed purchase outcome for that session.

A row represents a session, not necessarily one unique person forever. A person could have multiple sessions. These columns describe recorded behaviour; they do not prove a visitor's motivation.

3. Dataset column dictionary

Browsing behaviour columns

Column	Meaning
Administrative	Number of administrative pages visited during the session.
Administrative_Duration	Total time spent on administrative pages.
Informational	Number of informational pages visited during the session.
Informational_Duration	Total time spent on informational pages.
ProductRelated	Number of product-related pages visited during the session.
ProductRelated_Duration	Total time spent on product-related pages.
BounceRates	Analytics measure associated with visitors leaving after viewing a page without further interaction.
ExitRates	Analytics measure associated with pageviews that are the final pageview in a session.
PageValues	Analytics measure related to the value of pages viewed before a transaction.

These fields describe counts of pages or time spent during the session, along with analytics measures related to session navigation.

4. Dataset column dictionary

Session attributes and target column

Column	Meaning
SpecialDay	Numeric measure of how close the session date is to a special day, such as Valentine's Day or Mother's Day.
Month	Month in which the session occurred.
OperatingSystems	Category code identifying the visitor's operating system.
Browser	Category code identifying the visitor's browser.
Region	Geographic region category code.
TrafficType	Category code describing the traffic source/type.
VisitorType	Visitor category, commonly including Returning_Visitor, New_Visitor, and Other.
Weekend	Whether the session occurred on a weekend; Boolean in the original dataset.
Revenue	Target outcome: True if the session resulted in a purchase; False otherwise.

Important note about category codes

OperatingSystems, Browser, Region, and TrafficType are categorical codes. Their numeric values should not automatically be interpreted as quantities or rankings. VisitorType is a descriptive visitor category. Weekend indicates whether the session took place on a weekend.

Revenue is the target/outcome. When predicting Revenue, it should not be included among the input features for that same prediction.

5. Frequently asked dataset questions

Questions and answers for the RAG chatbot

Q: What does Revenue mean?

A: It records the purchase outcome for a session. True means a purchase occurred; False means it did not.

Q: What does ProductRelated mean?

A: It counts product-related pages visited during a session.

Q: What does ProductRelated_Duration mean?

A: It records the total time spent on product-related pages during a session.

Q: What is VisitorType?

A: It describes the visitor category, commonly including returning visitors, new visitors, and other.

Q: What does Weekend represent?

A: It indicates whether the session occurred on a weekend.

Q: What does PageValues mean?

A: It is an analytics measure related to the value of pages viewed before a transaction.

Q: Does each row represent one unique customer?

A: Not necessarily. Each row represents a browsing session, and a person could have multiple sessions.

Q: How many rows are in the original dataset?

A: The original dataset is documented as 12,330 sessions. The cleaned local project file previously contained 12,205 rows.

Q: Is this a product inventory dataset?

A: No. It focuses on browsing sessions and purchase intention/outcome, not stock or product inventory.

6. Dataset use and interpretation

Questions the dataset can help investigate

The recorded session features can be used to explore how page-visit counts, time spent on page categories, visitor type, month, and other session attributes relate to the Revenue outcome. Observed relationships should not automatically be interpreted as causes.

7. Important limitations

• The dataset describes browsing sessions, not a complete personal profile of each customer.

• Revenue is an observed target outcome; a prediction is an estimate, not a guarantee of purchase.

• Category codes do not necessarily represent an ordered scale.

• The knowledge base explains column meanings. It does not automatically calculate statistics from the current CSV.

• For exact row counts, averages, purchase rates, distributions, or charts, the Streamlit app must read the actual CSV and calculate those values.

8. Suggested RAG chatbot questions

• What does the Revenue column mean?

• Which columns describe browsing behaviour?

• What is the difference between ProductRelated and ProductRelated_Duration?

• What does VisitorType contain?

• Does each row represent a customer or a session?

• Which column is the target outcome?

• How many rows are in the original dataset versus the cleaned project file?

• Can you explain the Weekend and Month columns?

Recommended application design

Use this PDF as the RAG knowledge source for dataset-definition questions. Connect the actual online_shoppers_intention.csv file separately for dataset statistics and charts, and connect the trained model for purchase predictions. In Streamlit, keep these as related but distinct features: Dataset Chatbot, Dataset Analysis, and Purchase Prediction.